



Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads

Investigation source pack

Compiled 30/09/2026

INVESTIGATION WORKFLOW SUITE

Contents

Safety reports, master and deckhand.....	3
1. Master's report.....	3
2. Deckhand's report	4
Occurrence notification	6
The occurrence	6
Brief description	6
Immediate actions taken	7
Voyage data, AIS track and equipment logs	8
The channel, as charted	8
AIS track, inbound.....	8
What the track shows	9
Engine monitoring recorder.....	9
Chart plotter internal event log	10
Echo sounder alarm log.....	10
VHF and bridge audio extract	11
Tide, weather and light	14
Tide.....	14
Weather	14
Light and moon.....	14
Plain reading.....	15
Operator and port procedure extracts.....	16
1. Waterways Aids to Navigation Code WAN-2, section 4, "Availability and defect reporting"	16
2. Aids to navigation services agreement, Kanooka Shire Council and Kanooka Heads Port Authority, schedule 2	16
3. Tarrakin Ferries Operations Manual OPS-3, "Passage planning", revision 6, 02/2023.....	17
4. Tarrakin Ferries Operations Manual OPS-9, "Defect reporting and rectification", revision 4, 08/2024.....	18
5. Route Risk Assessment RRA-07, Weerona Channel, issue 2, 03/2023	18
6. Change record CHG-2026-014, "Winter timetable, Muntari Island route", 04/2026.....	18
Vessel, port and crewing records.....	20
1. Vessel maintenance history and defect book, MV Tarrakin Star	20
2. Post-grounding survey and slipping, 09/07/2026 to 14/07/2026	20
3. Lifejacket stowage survey, 16/07/2026.....	21
4. Crew records and roster.....	21
5. Kanooka Heads Port Authority aids to navigation records, No. 4 beacon	22
6. Kanooka Heads foreshore lighting survey, 18/07/2026.....	23
Interview transcript, master.....	24
Interview notes, deckhand and outgoing master	26
1. Deckhand, 14/07/2026, 09:40, Tarrakin Ferries town wharf office	26
2. Outgoing master, morning shift 07/07/2026, interviewed 17/07/2026, 11:15, by telephone	28

Safety reports, master and deckhand

The master's report, written from the vessel forty minutes after the grounding and the first written account of it, and the deckhand's, submitted two days later about something nobody had asked her about. Both are quoted exactly as they were written.

Fictional source material for the Kanooka Heads case study. Submitted by role; no name appears in the investigation file.

Two reports were submitted. The master's is the first written account of the occurrence and was in the safety system before the operator's own notification was raised, so it, rather than the notification, is what opened this investigation. The deckhand's was submitted two days later and raised something nobody had asked about.

1. Master's report

Report number: SR-2026-0288

Submitted by: Master, MV Tarrakin Star, 07/07/2026

Submitted: 07/07/2026 08:52 UTC (18:52 local), from the vessel

Scheme: Submitted under the company's just culture reporting policy. The report may inform the investigation; it is not to be quoted into a distributed document.

Text as submitted:

TF18 Muntari to Kanooka Heads, 7 July, last service. Thirty eight passengers and three crew. Left Muntari on time at 17:35. Dark, clear, light westerly, flat water, nothing in the channel.

I entered Weerona from the north on the centreline and I was happy with where I was at number two. About a minute and a half after that the plotter froze. Picture stayed on the screen, boat stopped moving on it, wouldn't take a touch. I tried it twice. It had done the same thing to the morning bloke and he'd told me about it at changeover, so I wasn't surprised by it, which is probably the thing I'd most like to take back.

I should have stopped there. There is nothing wrong with stopping in that channel on a flat night with no other traffic, and if you'd asked me a week ago what I'd do I'd have told you I'd stop. I didn't. I kept going at eight knots and went looking for number four with my eyes, because that's the transit and I've done it four thousand times.

I couldn't find it. I looked for it for what felt like a long time and I know now it was about a minute and a half. There is a lot of light behind that beacon now with the new car park up, and a two and a half second green flash

in front of all that is not what it used to be. When I finally saw it, it was nowhere near where I had it in my head, and I understood straight away I was well to the west. I had already put her over five degrees looking for it without really deciding to.

The shallow alarm went and I had both throttles astern within a few seconds and she hit anyway. I stopped the engines. I did not try to back her off. The tide was falling and I did not know what I had under me, and I would rather have a boat sitting on a bank with everyone on it than a boat in deep water with a hole in her. I got jackets on everyone, got the void sounded, told Port Control, and the workboat had them off inside an hour.

Two things I want on the record.

The first is mine. I was told about that plotter and I did not put it in the book, and I did not stop when it went. Nobody made me do either of those things.

The second is not mine. That beacon has been getting harder to see for two years and every one of us has said so to each other and none of us has ever written it down anywhere, because there is nowhere to write it down. We do not have a night plan for that channel. We were never given one. The timetable moved into the dark in June and nothing else about the run changed at all. I would like somebody to look at that, because the next bloke will be me on a worse night.

2. Deckhand's report

Report number: SR-2026-0291

Submitted by: Deckhand, MV Tarrakin Star, 07/07/2026

Submitted: 09/07/2026 23:10 UTC (10/07/2026 09:10 local)

Scheme: Submitted under the company's just culture reporting policy.

Text as submitted:

I want to report something about the lifejackets in the forward saloon because it has been bothering me since Tuesday night.

When the master told me to get jackets on everyone I started at the back and worked forward. The aft saloon is fine, the locker is on the bulkhead and people were opening it themselves before I got there. The forward saloon is not fine. That locker is behind the kiosk counter. There were three people standing at the counter, which is where people stand, and I had to ask them to move and then go round the end of the counter through a gap you can just about get a shoulder through, and hand jackets out over the counter one at a

time.

It took me a couple of minutes and everyone was calm and it did not matter on Tuesday. It would have mattered on a different night. The sign on the wall in there says the jackets are in the locker forward on the port side and you cannot see that locker from any of the seats, so nobody in that saloon could have got one without me.

I am not raising this about the grounding. I am raising it because I think it is the same on the other two boats and I would not know how to get forty jackets out of there in a hurry.

Occurrence notification

The notification raised by the operations manager that evening: what happened, where, the damage, the injuries, and the actions taken straight away.

Fictional. Tarrakin Ferries, the MV Tarrakin Star, Kanooka Heads, Muntari Island, the Weerona Channel, Boolarong Bank, the people and every manual, standard, record and reference number in this file are invented for teaching purposes. Any resemblance to a real operator, port, vessel, waterway or occurrence is coincidental.

Notification reference: TFP-OCC-2026-0093

Raised by: Operations Manager, Tarrakin Ferries Pty Ltd

Raised at: 07/07/2026 09:05 UTC (19:05 local)

Initial risk rating: High

Investigation required: Yes (risk rating Medium or above)

The occurrence

Item	Detail
Date	07/07/2026
Time	08:12:04 UTC (18:12:04 local, AEST UTC+10)
Location	Weerona Channel, Kanooka River entrance, far north coast of New South Wales
Position	54 m west of the dredged channel's western edge, 34 m west of the No. 4 starboard-hand beacon
Operator	Tarrakin Ferries Pty Ltd
Vessel	MV Tarrakin Star, 24.4 m aluminium catamaran passenger ferry, built 2011, survey class 1D, 96 passengers certificated
Voyage	Service TF18, Muntari Island to Kanooka Heads Town Wharf, the last service of the day
Persons on board	38 passengers, 1 master, 1 deckhand, 1 kiosk attendant
Phase	Channel transit, inbound, in darkness
Injuries	Nil. One passenger reported a grazed shin; two passengers were treated at the wharf for distress.
Damage	Substantial. Starboard hull bottom plating set up over 4.1 m with a 180 mm split in way of the No. 2 void; starboard propeller and shaft bent; starboard rudder stock distorted; starboard skeg torn away.
Third party	Nil. Approximately 40 m of the Boolarong Bank seagrass margin disturbed. Weerona Channel closed to commercial traffic for 19 hours.

Brief description

TF18 was the last service of the day, one of the winter timetable's evening returns, which since 01/06/2026 have been made in darkness. It departed Muntari Island at 17:35 local with 38 passengers embarked. The end of civil twilight was 17:29 and the channel transit was made in full darkness.

The vessel entered the Weerona Channel from the north at about 18:04 and, on

the master's account and on the recorded AIS track, was on the charted centreline abeam the No. 2 beacon. At about 18:07 the bridge chart plotter display froze. The master continued the transit on visual reference, looking for the No. 4 starboard-hand beacon on the western side of the channel. The vessel set progressively to starboard. The echo sounder's shallow alarm sounded at 18:11:48 and the master ordered both engines astern four seconds later. At 18:12:04 the vessel made contact with Boolarong Bank at 6.8 kt over the ground and came to rest within 11 m, heeled 4 degrees to starboard and hard aground on a falling tide.

The master stopped both engines and did not attempt to refloat. Passengers were mustered and lifejackets issued in both saloons, and the vessel's position was passed to Kanooka Heads Port Control on VHF channel 16. The port workboat came alongside at 18:41 and all 38 passengers were transferred ashore by 19:26. The vessel was refloated on the rising tide at 00:20 on 08/07/2026 and towed to the town wharf, where a starboard hull breach and progressive flooding of the No. 2 void were confirmed.

Immediate actions taken

- Weerona Channel closed to commercial traffic by the Harbour Master pending survey of the grounding position and recovery of the vessel.
- Vessel withdrawn from service and slipped at Kanooka Heads on 09/07/2026. Chart plotter and echo sounder heads removed and sealed for examination.
- Master and deckhand stood down from duty and tested for drugs and alcohol; both results negative.
- Evening return services on the Muntari Island route suspended by the General Manager Operations pending the outcome of this investigation.

Voyage data, AIS track and equipment logs

The shore AIS receiver's track of the transit, the engine and steering recorder, and the chart plotter's and echo sounder's own logs, which between them fix where the vessel was, what was being done with it, and when the plotter stopped.

Fictional source material for the Kanooka Heads case study.

Source: Shore-based AIS receiver log held by Kanooka Heads Port Control; the vessel's engine monitoring recorder; the chart plotter's internal event log; the echo sounder's alarm log.

Prepared by: Lead investigator with the Port Authority's traffic officer.

Reference: AIS-2026-0093-R1, issued 15/07/2026.

Clock: AIS timestamps are UTC from the receiver. The plotter and echo sounder logs were found 2 seconds fast against the AIS receiver and have been corrected by that amount throughout.

The channel, as charted

The Weerona Channel is dredged to 3.2 m below chart datum, 40 m wide, and runs 195 degrees true inbound over a length of 1.35 nm between the No. 1 and No. 6 beacons. Boolarong Bank lies along its western side and dries in places to 0.6 m above chart datum. The No. 4 starboard-hand beacon stands on the eastern edge of the bank, 20 m west of the channel's western edge, and is the only lit mark between the No. 2 beacon and the channel's southern end. The No. 1 beacon marks the channel's northern entrance and stands 1.13 nm north of the No. 4; the No. 2 is 0.90 nm north of it. Both are on the eastern side of the channel.

The vessel's draught on the night was 1.35 m.

AIS track, inbound

UTC	Local	Offset from centreline	North of No. 4	Speed over ground	Course over ground	Note
08:03:58	18:03:58	2 m east	1.13 nm	8.4 kt	195	Abeam No. 1 beacon, entering the channel
08:05:30	18:05:30	1 m west	0.90 nm	8.4 kt	195	Abeam No. 2 beacon, on the centreline
08:07:06	18:07:06	3 m west	0.68 nm	8.4 kt	196	Chart plotter event log records a display freeze at this time
08:08:00	18:08:00	7 m west	0.54 nm	8.1 kt	198	
08:08:31	18:08:31	12 m west	0.47 nm	8.0 kt	199	Bridge audio: "Right. Where's four"
08:09:00	18:09:00	20 m west	0.40 nm	8.0 kt	199	Vessel now at the channel's western edge
08:10:02	18:10:02	39 m west	0.25 nm	7.9 kt	200	Bridge audio: "There's four"; outside the dredged channel

08:11:00	18:11:00	58 m west	0.13 nm	7.6 kt	199	Over Boolarong Bank
08:11:48	18:11:48	70 m west	0.03 nm	7.1 kt	198	Echo sounder shallow alarm, 0.9 m under keel
08:11:52	18:11:52	71 m west	--	7.0 kt	198	Both engine throttles to astern
08:12:04	18:12:04	74 m west	abeam	6.8 kt	197	Contact. Longitudinal deceleration spike; speed to zero in 11 m
08:12:31	18:12:31	--	--	0 kt	--	Both engines stopped

The grounding position was surveyed on 08/07/2026 at 74 m west of the charted channel centreline, which is 54 m west of the channel's western edge and 34 m west of the No. 4 beacon.

What the track shows

The vessel was on the charted centreline abeam the No. 2 beacon and held it until 08:07:06. From that time the heading increased steadily from 195 to 200 degrees and the vessel set west at an average rate of 12 m per minute, crossing the channel's western edge at about 08:09:00 and reaching the bank at about 08:11:00.

The set is not a current effect. It is a continuous heading change of five degrees applied over about a minute and then held, consistent with a vessel steered onto a new heading and kept there. The recorded rate of turn peaks at 2.1 degrees per minute at 08:07:40 and returns to zero by 08:08:50.

Speed over the ground fell from 8.4 kt to 7.6 kt between 08:07 and 08:11 with no throttle movement recorded, consistent with the increasing drag of shoaling water beneath the hulls.

The distance north of the No. 4 beacon is given at each point because the beacon's own visibility varies with it. The vessel was 1.13 nm north of the beacon entering the channel, 0.90 nm abeam the No. 2, 0.68 nm when the plotter froze, 0.47 nm when the master began looking for it, and 0.25 nm when he called it.

Engine monitoring recorder

Both engines ran at 1,780 and 1,784 rpm from 08:03 to 08:11:52, within 12 rpm of each other throughout, with no alarm, no temperature or pressure excursion and no throttle movement in that period. Steering gear hydraulic pressure is recorded continuously at 96 to 99 bar against a normal band of 90 to 105 bar, with no loss at any point. Rudder angle follows helm demand within 1.5 degrees throughout the transit.

At 08:11:52 both throttles move to the astern detent together. Astern thrust is developing at the moment of contact but had not reached full.

Chart plotter internal event log

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued 03/2024.

Log time (corrected)	Entry
07/07/2026 00:42 UTC	Watchdog reset, cause code 0x51 (display process not responding)
07/07/2026 01:17 UTC	Watchdog reset, cause code 0x51
07/07/2026 08:07:06 UTC	Display process not responding; no reset recorded
07/07/2026 08:16 UTC	Power removed

The two entries at 00:42 and 01:17 UTC correspond to 10:42 and 11:17 local on the morning of 07/07/2026, during the morning master's shift. The 08:07:06 entry has no accompanying reset: on that occasion the display process stopped responding and the unit did not recover itself.

The unit was bench tested by an approved electronics contractor on 21/07/2026 over 46 hours with the same antenna and cabling. The fault did not recur and no defect was found. The contractor's report notes that cause code 0x51 is known to the manufacturer and is addressed by firmware issued in 11/2024, which had not been applied to this unit.

Echo sounder alarm log

Shallow alarm set to 0.9 m under the keel. This is the value written into the setup when the unit was commissioned in 2019 and it has not been changed. The alarm sounded once on the voyage, at 08:11:48 UTC, 16 seconds before contact. At 7.1 kt over the ground the vessel covered 58 m in those 16 seconds. At the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting. The vessel's own trials figure for a crash stop from 8 kt is 92 m.

VHF and bridge audio extract

A verbatim transcript of both VHF channels and the vessel's own bridge camera audio, from the departure call to the last passenger ashore.

Fictional source material for the Kanooka Heads case study. Speakers are identified by role only.

Sources: Kanooka Heads Port Control recorded VHF channel 12 (port operations) and channel 16; and the audio track of the vessel's own bridge camera, which records continuously to a 30-day loop and was downloaded before the vessel was slipped.

Reference: AUD-2026-0093, transcribed 16/07/2026 by the investigator with the Port Authority's traffic officer verifying. Times are UTC, corrected to the AIS receiver clock.

Record type: Verbatim. Words in square brackets are the transcriber's note, not speech. Three dots mark a pause; "(unintelligible)" marks audio that two listeners could not agree on.

MAS is the master. DEC is the deckhand. KIO is the kiosk attendant. PC is Kanooka Heads Port Control.

...

07:32:10 MAS [Public address] Good evening everyone, welcome aboard, we're about twenty five minutes across to the town wharf tonight.

It's dark and it's a bit fresh out on the aft deck, so if you'd rather be inside there's plenty of room.

07:35:02 MAS [VHF 12] Port Control, Tarrakin Star, departing Muntari, one eight for Kanooka Heads, thirty eight passengers.

07:35:11 PC Tarrakin Star, Port Control, copied, nothing to affect you, channel's clear.

08:03:44 MAS [Bridge] Number one's coming up. Right, that's us in.

08:04:10 DEC Want me to stay up?

08:04:14 MAS Nah, go and do the fares, I'll give you a yell coming up on the wharf.

08:05:33 MAS [Bridge, to himself] Two ... good.

08:07:11 MAS Oh, come on.

08:07:26 MAS [Sound consistent with repeated screen taps.]

08:07:44 MAS (unintelligible) frozen again.

08:08:31 MAS Right. Where's four.

08:09:40 MAS [Sound consistent with a window or door being slid open.]

08:10:02 MAS [To himself] There's four. Keep her over.

08:11:48 [Shallow alarm tone begins. Continues to 08:12:20.]

08:11:50 MAS No no no.

08:11:52 [Sound consistent with both throttles being moved together.]

08:12:04 [Heavy impact. Sustained grinding and vibration for approximately 9 seconds.]

08:12:16 KIO (unintelligible) is everyone alright back there?

08:12:31 MAS [Sound consistent with engine stop controls being operated.]

08:12:40 MAS [Public address] Ladies and gentlemen, this is the master. We have touched the bottom. The vessel is stopped and she is safe. Please stay in your seats, my crew are coming round to you now, and I'll be back to you in one minute.

08:13:04 MAS [VHF 16] Pan pan, pan pan, pan pan, Kanooka Heads Port Control, Tarrakin Star. We are aground on Boolarong Bank, west side of the Weerona Channel, abeam number four. Thirty eight passengers, three crew, no injuries reported, no pollution. Vessel is stable, we are upright, I am not attempting to refloat. Request the workboat for a passenger transfer.

08:13:29 PC Tarrakin Star, Port Control, copied your pan pan, aground abeam number four, thirty eight passengers, three crew, no injuries, not refloating. Workboat is being called now. Confirm you are not taking water?

08:13:47 MAS Port Control, Tarrakin Star, deckhand is sounding the voids now, I'll come back to you. Be aware the tide is falling, she will sit down further.

08:14:52 MAS [Public address] Ladies and gentlemen, the master again. We have run aground in the channel. Nobody is hurt and the vessel is not in danger. My crew are handing out lifejackets now and I would like everyone to put one on and keep it on. We are not going into the water. A boat is coming out from the town to take you off and it will be about half an hour.

08:16:20 DEC Number two void's got water in it. Not fast. Two, maybe three inches on the sounding rod.

08:16:31 MAS Right. Keep sounding it every ten minutes and tell me if it moves.

08:17:05 MAS [VHF 16] Port Control, Tarrakin Star. Slow ingress into the

number two starboard void, approximately seventy five millimetres on the sounding rod, being monitored ten minutely.

No ingress anywhere else. No change to my intentions.

08:22:40 DEC That's everyone in a jacket. There's two in the forward saloon I had to go round the servery for, the locker's behind the counter and there's people standing at it.

08:22:51 MAS Yeah. Write that one down.

08:41:15 PC Tarrakin Star, Port Control, workboat is coming alongside your port quarter now.

09:26:08 MAS [VHF 16] Port Control, Tarrakin Star, all thirty eight passengers ashore, crew of three remaining on board.

'''

Transcriber's note. No transmission on either channel between 07:35 and 08:13 concerns the No. 4 beacon, the channel marks or the plotter. Port Control's recorded traffic for the 24 hours before the occurrence contains no report of a defective aid to navigation. The master's remark at 08:08:31, "Right. Where's four", and at 08:10:02, "There's four", are 91 seconds apart, and the vessel crossed the channel's western edge between them.

Tide, weather and light

The tide predictions and the port's gauge record, the weather observations, and the light and moon data for the night, with the investigator's plain reading of what they do and do not explain.

Fictional source material for the Kanooka Heads case study.

Source: Kanooka Heads Port Authority tide gauge archive, the published tide predictions for the Kanooka River entrance, the coastal weather observations for Kanooka Heads, and the astronomical data for the position.

Reference: MET-TIDE-0093-070726, retrieved 09/07/2026.

Record type: Verbatim copies of the predictions, the gauge record and the observations, with the investigator's plain reading at the end.

Tide

...

Kanooka River entrance, 07/07/2026 (AEST, UTC+10)

Predicted high water 15:20 1.66 m above chart datum

Predicted low water 21:34 0.30 m above chart datum

Predicted height at 18:12 0.94 m

Port Authority gauge, Kanooka Heads (same datum)

Recorded height at 18:00 0.99 m

Recorded height at 18:12 0.91 m

Recorded height at 18:30 0.79 m

Largest departure from prediction in the 12 hours to 20:00: 0.07 m at 11:40

...

Weather

...

Kanooka Heads AWS, 07/07/2026 (AEST)

17:30 Wind 310 degrees 9 kt Vis >10 km Cloud FEW 3,500 ft Temp 14 QNH 1021

18:00 Wind 305 degrees 8 kt Vis >10 km Cloud FEW 3,500 ft Temp 13 QNH 1021

18:30 Wind 310 degrees 7 kt Vis >10 km Cloud SCT 3,500 ft Temp 13 QNH 1022

Rainfall since 09:00: 0.0 mm

Sea state in the channel, reported by the port workboat at 18:45: slight, wind ripple only, no swell penetrating the entrance.

...

Light and moon

...

Kanooka Heads, 07/07/2026 (AEST, UTC+10)

Sunset 17:02
End of civil twilight 17:29
Moonrise 06:12 (waning crescent, 2% illuminated)
...

Plain reading

The weather was not a factor and the tide was not a factor. The wind was offshore at 7 to 9 kt with no sea in the channel, the visibility was better than 10 km with no precipitation and no fog, and the port workboat crew who came out at 18:41 described the conditions as flat.

The tide matters only for what it says about depth, and it says that the depth on the night was the depth the chart gives. The gauge held within 0.07 m of the prediction all day and was within 0.03 m of it at the moment of the grounding. There was no surge, no set-down and no wind effect on the water level of any consequence. The vessel grounded in the water depth a navigator reading the chart and the tide table would have expected.

What that depth was, at the position surveyed on 08/07/2026: the bed at the contact point is 0.50 m below chart datum, so with the gauge at 0.91 m there was 1.41 m of water. The vessel's static draught was 1.35 m. Squat for this hull form at 6.8 kt in that depth of water is calculated at 0.18 m, giving an effective draught of 1.53 m against 1.41 m of water. The vessel touched because it was outside the dredged channel, not because the water was lower than it should have been.

The light condition is the one thing in this record the investigation has to carry forward. The transit began 35 minutes after the end of civil twilight with the moon not due to rise until the following morning and 2% illuminated when it did. There was no natural light on the water at all. Every reference available to the master on the last two thirds of the channel was an artificial light: the channel beacons, the town, and the new foreshore lighting behind them.

Operator and port procedure extracts

Verbatim extracts from the aids to navigation code, the port services agreement, the operator's passage planning and defect reporting manuals, the route risk assessment and the change record, transcribed from the copies current on the day.

Fictional source material for the Kanooka Heads case study. Every document, reference number and revision date below is invented.

Collected by: Lead investigator, 10/07/2026 to 20/07/2026.

Record type: Verbatim extracts, transcribed from the controlled copies current on 07/07/2026.

1. Waterways Aids to Navigation Code WAN-2, section 4, "Availability and defect reporting"

4.1 The authority responsible for an aid to navigation must maintain it so that it performs the function for which it was established.

4.2 Where an aid to navigation is established on a declared shipping channel, the responsible authority must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect.

4.3 Where an aid to navigation is established on a waterway that is not a declared shipping channel, section 4.2 does not apply. The responsible authority must be able to demonstrate that the aid is inspected at intervals it has determined and recorded.

The Weerona Channel is not a declared shipping channel. It is a dredged recreational and commercial access channel on a waterway managed by Kanooka Shire Council through the Kanooka Heads Port Authority. Section 4.2 accordingly does not apply to any of its six beacons. There is no availability target for them, no obligation to record availability, and no obligation to notify any change to their character or range to the operators who use the channel. The Authority is not in breach of the code.

2. Aids to navigation services agreement, Kanooka Shire Council and Kanooka Heads Port Authority, schedule 2

S2.1 The Authority will inspect each lit aid by daylight once per quarter and will confirm each lit aid's operation by observation from the town wharf once per month.

S2.4 Vegetation management within 50 m of an aid is a works item and is funded from the general waterways maintenance vote. Vegetation beyond 50 m

is not within the scope of this agreement.

S2.7 Defects found are recorded in the aids to navigation register and actioned in accordance with the Authority's maintenance priority scale.

There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid.

The monthly confirmation at S2.1 is made from the town wharf, which is 2.1 nm south of the No. 4 beacon and on the beacon's southern side. Nothing in the agreement requires any aid to be observed from the approach a vessel actually uses.

3. Tarrakin Ferries Operations Manual OPS-3, "Passage planning", revision 6, 02/2023

3.2 A written passage plan is held for each scheduled route. The master will conduct the passage in accordance with the plan and will record any departure from it in the deck log.

3.4 The Weerona Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon.

3.6 In reduced visibility the master will reduce to a speed at which the vessel can be stopped within the distance in which the channel can be seen, and will consider anchoring or returning to open water.

OPS-3 Appendix A, Weerona Channel passage plan, issue 2, 02/2023.

Transit speed 8 knots. Course 195 degrees true inbound. Primary means of position fixing: chart plotter. Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights. Aids: No. 1 (Fl G 4s, range 4 M), No. 2 (Fl R 4s, range 4 M), No. 4 (Fl G 2.5s, range 6 M), No. 6 (Fl R 2.5s, range 4 M). Hazards: Boolarong Bank to the west of the channel, drying. Minimum depth 3.2 m below datum. Note: this plan is written for the daylight transit. **Night transits are not scheduled on this route.**

The appendix is issue 2, dated 02/2023, and is the current issue. The last line is the plan's only reference to darkness. There is no night passage plan for the Weerona Channel, no guidance on what to do if the plotter is lost during the transit, and no abort or hold point named anywhere in the plan. The town wharf leading lights named as the secondary means bear 195 degrees but are 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third.

4. Tarrakin Ferries Operations Manual OPS-9, "Defect reporting and rectification", revision 4, 08/2024

9.1 A defect which renders the vessel unseaworthy or which affects a statutory item of equipment shall be reported to the Operations Manager immediately and entered in the vessel defect book.

9.2 All other defects shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the Operations Manager weekly.

9.3 At a change of master the outgoing master will brief the relieving master on the state of the vessel, any defect, and anything else affecting the conduct of the service.

Nothing in OPS-9 requires the change-of-master brief at 9.3 to be written down, and nothing requires the relieving master to confirm that a defect mentioned in it has been entered in the defect book. The defect book is a bound book kept in the wheelhouse and is transcribed into the maintenance system when the Operations Manager reviews it. An intermittent fault on a non-statutory item falls under 9.2, so it is due at the end of the shift in which it is found, and no other trigger exists for it.

The chart plotter is not a statutory item on a class 1D vessel on this route. It is the primary means of position fixing named in the passage plan.

5. Route Risk Assessment RRA-07, Weerona Channel, issue 2, 03/2023

Hazards assessed: Boolarong Bank to the west of the channel; channel width 40 m against a vessel beam of 8.2 m; single unlit reach between the No. 2 and No. 4 beacons; tidal range 1.4 m; recreational traffic at weekends.

Controls: transit at 8 knots on the charted centreline; chart plotter monitored throughout; No. 4 beacon identified before passing No. 1.

Operating restriction: daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan.

Issue 2 is the current issue. No issue 3 exists.

6. Change record CHG-2026-014, "Winter timetable, Muntari Island route", 04/2026

Scope of change: winter timetable from 01/06/2026 to 31/08/2026. The last return service TF18 moves from 16:15 to 17:35 to meet the afternoon coach connection at the town wharf.

Assessment: timetable change only. Existing route, existing vessel, existing crew qualifications, existing berths. No new hazard identified.

Route risk assessment review required: No.

Approved: Operations Manager, 22/04/2026.

From 01/06/2026 the TF18 transit of the Weerona Channel is made after the end of civil twilight on every day of the winter timetable. The daylight-only restriction at RRA-07 issue 2 is not referenced anywhere in CHG-2026-014, RRA-07 was not reopened, no night passage plan was written, and the General Manager Operations' approval required by RRA-07 for a night transit was neither sought nor given. The change was approved one level below the person whose approval the restriction names.

TF18 ran 31 times in darkness between 01/06/2026 and 07/07/2026.

Vessel, port and crewing records

The investigator's description of what the vessel, survey, stowage, crewing and Port Authority records show, with the register's own wording quoted where it matters.

Fictional source material for the Kanooka Heads case study.

Collected by: Lead investigator, 09/07/2026 to 24/07/2026.

Record type: Investigator's description of what the records show, except where an extract is quoted.

1. Vessel maintenance history and defect book, MV Tarrakin Star

The vessel completed its annual survey on 11/03/2026 with no outstanding conditions. The last slipping was 03/2025.

The wheelhouse defect book carries eleven entries in the 90 days before the occurrence. None concerns the chart plotter, the GPS antenna, the echo sounder, the steering gear, the engines or the propulsion controls. The open items on 07/07/2026 were a seized port aft mooring bollard cover, a saloon window seal weeping and a defective deck light on the aft boarding gate.

There is no entry of any kind for 07/07/2026 before the occurrence. The morning master's shift ended at 13:05 local. The plotter's own event log records display freezes at 10:42 and 11:17 local that morning.

The maintenance system holds no work order, no fault report and no correspondence relating to the chart plotter at any time in 2026. The firmware bulletin the electronics contractor identified (issued 11/2024, addressing cause code 0x51) was not held by the operator and no subscription to the manufacturer's bulletins existed.

2. Post-grounding survey and slipping, 09/07/2026 to 14/07/2026

- Starboard hull: bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in the plating on the centreline of the hull in way of the No. 2 void. Ingress into the No. 2 void confirmed; the deckhand's sounding rod gave approximately 75 mm at 18:16 and approximately 260 mm when the vessel was refloated at 00:20.
- Starboard propeller: all three blades bent aft, tips distorted. Shaft bent 1.4 mm over its length. Stern gland undamaged.
- Starboard rudder stock distorted 3 degrees; starboard skeg torn away and recovered from the bank on 08/07/2026.
- Port hull, propulsion and steering: no damage found.
- Steering gear: both rams, tiller arms, the hydraulic system and the autopilot removed and function tested on 12/07/2026. All serviceable. No

internal leakage, no air in the system, no sign of any pre-existing defect. The system passed a full lock-to-lock timing test inside the specified period.

- Grounding position survey, 08/07/2026: bed level at the contact point 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west. The bank gradient across the contact section is about 1 in 70. Furrows in the bed consistent with the starboard hull and the skeg run 11 m from the point of first contact on a heading of 197 degrees.

3. Lifejacket stowage survey, 16/07/2026

Carried out following the deckhand's account and safety report.

The vessel carries 120 adult and 20 child lifejackets in four lockers. The forward saloon's locker holds 40 adult jackets and is mounted in the bulkhead behind the kiosk servery counter. The counter was fitted during a 2019 interior refit. The arrangement approved at build in 2011 had an open bench in that position with the locker face clear.

Measured on 16/07/2026: the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the servery counter through a 0.6 m gap at its inboard end. With three people standing at the counter, as there were on the night, that gap is the only route. The muster placard in the forward saloon directs passengers to "the locker forward on the port side", which is the locker in question and is not visible from the seats.

The 2019 refit was carried out under a minor modification notice and did not trigger a re-survey of the safety equipment arrangement. The same servery counter is fitted to the operator's two other 24 m vessels.

4. Crew records and roster

Master. Certificate of competency, master less than 35 m, held since 2012. With Tarrakin Ferries 11 years, on the Muntari Island route 11 years. Approximately 4,100 transits of the Weerona Channel. Last route check 14/02/2026, satisfactory. No prior occurrence history. Had made 26 of the 31 darkness transits of the channel since the winter timetable began on 01/06/2026.

Deckhand. With the operator 7 months. Certificate of safety training and coxswain grade 2. First winter season on the route.

Outgoing master (morning shift, 07/07/2026). With Tarrakin Ferries 6 years, on the route 4 years.

Roster and duty, 07/07/2026. The route is worked on a 12 days on,

2 days off cycle. 07/07/2026 was the master's ninth consecutive day. The day's duty was a split: 05:40 to 08:40 local and 13:10 to 19:20 local, giving 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes, with a 4 hour 30 minute unpaid break at home mid-day. Both figures are inside the operator's own limits (10 hours of duty, 14 hours of spread) and inside the state's crewing determination. The master recorded 6 hours 15 minutes of sleep on the night of 06/07 in the fatigue diary the operator asks masters to keep but does not collect.

5. Kanooka Heads Port Authority aids to navigation records, No. 4 beacon

Solar conversion, 11/2025. The No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character (Fl G 2.5s) did not change. A notification was lodged with the hydrographic service on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still records the No. 4 beacon at 6 M.

Range and visibility check, 18/07/2026. Carried out at the investigator's request, from the port workboat, on the inbound track and at the same time of night. The No. 4 beacon's lantern was measured at a luminous range of 2.6 nm against its nominal 3. More to the point, the light is not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon: the sightline over that stretch passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. It is visible beyond 0.84 nm, where the sightline passes clear of the islet's northern edge, and again inside 0.34 nm, where it opens past its southern edge. The beacon's lantern is 4.2 m above chart datum. The islet's canopy was measured at 5.8 m; the aerial survey flown for the Council in 2019 gives it as 2.1 m.

Aids to navigation register, vegetation items. Work order WO-118, raised 09/2023: "Mangrove regrowth, Weerona Islet, screening No. 4 beacon from the northern approach between approximately half a mile and a quarter mile. Recommend trimming to 2 m and a three-yearly cycle thereafter. Estimated \$18,500." Deferred at the 2023, 2024 and 2025 works rounds, each entry reading "defer, unfunded, refer next round". Work order WO-176, raised 03/2026, records the same sightline as "now fully closed on the northern

approach". The Authority's file note against WO-176 records that the islet is 240 m from the beacon and so falls outside the 50 m scope of schedule 2.4 of the services agreement, and that the work therefore competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

Monthly observation log. The No. 4 beacon is recorded as "observed, operating" on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The check is made from the town wharf, 2.1 nm south of the beacon, from which Weerona Islet is not in the line of sight.

Defect and notification history. No defect has ever been recorded against the No. 4 beacon. No notice about it has ever been issued to a waterway user. The Authority holds no record of any operator asking about the channel's aids, and the aids to navigation register has never been provided to Tarrakin Ferries.

6. Kanooka Heads foreshore lighting survey, 18/07/2026

The Kanooka Heads foreshore redevelopment was commissioned on 22/05/2026, four weeks after the winter timetable was approved and ten days before it began.

It includes 14 LED floodlights on 12 m columns lighting the new boat ramp and car park, 300 m east of the town wharf.

Measured from the port workboat on the inbound track: each floodlight presents about 4,800 candela toward the channel and none is shielded to seaward. From 0.5 nm north of the No. 4 beacon the floodlight array lies behind the beacon within 4 degrees of the same bearing and within 1.5 degrees of the same elevation, and the array is continuously lit while the beacon flashes green for 0.4 seconds every 2.5 seconds.

The redevelopment was approved by the Council's planning unit. The Port Authority's aids to navigation officer was not consulted, and no assessment of the lighting against the channel's aids was carried out or required.

Interview transcript, master

The Teams transcript of the recorded interview with the master, with every speaker identified by role.

Lead investigator: We're recording. Thanks for coming in. Just for the record, you're happy for us to record this?

Master: Yeah, go ahead.

Lead investigator: Good. What I'd like first is for you to take me through Tuesday in your own words. Start wherever you want and don't worry about what I might be after. I'll keep my questions until you're finished.

Master: Right. So it was day nine of a twelve, and it was a split, which is what we do in winter. Early run, home in the middle of the day, back on at ten past one. I took her over from the morning bloke about five past one at the town wharf. Nothing out of the ordinary. He told me the plotter had frozen on him twice that morning and come good on its own both times. I said something like, yeah, it did that to me last month. And that was the end of it. Neither of us wrote it down. The afternoon was normal. Two round trips. Then TF18 is the last one, seventeen thirty five off Muntari. Thirty eight on. It was properly dark by then. Clear, bit of westerly, no sea in the entrance at all. Best conditions you could ask for, honestly. Came down and picked up number one and went into the channel on the line. At number two I was happy. I was where I wanted to be. And then the plotter froze. The picture stays there, that's the thing, it looks fine. The boat just stops moving on it. And it wouldn't take a touch. I had a go at it twice. Then I went looking for number four out the window, because that's what you do. Centreline, plotter, number four. That's the transit. I've done it four thousand times. And it wasn't there. I couldn't find it. I opened the window because I thought maybe the glass, and it still wasn't there. When I did pick it up it was dead ahead of me instead of out on the starboard bow where it belongs, and I knew immediately. Because if four is dead ahead, I'm on top of the bank. The alarm went about a minute after that. I had both of them astern straight away and she went on anyway. Stopped the engines. Didn't touch them again. Got on the box to Port Control, got the jackets out, got the void sounded, and waited for the workboat. That's it, really.

Lead investigator: Thank you. I'd like to go back to the changeover. What exactly did he say to you about the plotter?

Master: He said it froze on him twice on the morning runs and it came back by itself both times. He said it was probably the heat in the wheelhouse. He wasn't worried about it and neither was I.

Lead investigator: Did either of you enter it in the defect book?

Master: No. And I've thought about that more than anything else. The book's a foot from where I was standing. It would have taken eleven seconds. But it's not a defect in your head, is it. It's a thing that happened and stopped happening. If it had stayed frozen he'd have written it up and I'd have had a different boat.

Second investigator: Is that how it usually goes on the boats? A thing that comes good doesn't get written up?

Master: Honestly, yes. The book is for things that are broken when you hand over. If it's broken you write it, if it came good you say it. Everyone does that and nobody has ever told us not to.

Lead investigator: When the plotter froze in the channel, did you consider stopping?

Master: No. And that's the honest answer and it's not a good one. I didn't think about it. I went straight to the visual transit because in my head the visual transit is the backup. There's nothing

that tells you to stop. There's nothing that tells you anything. We haven't got a night plan for that channel. The plan we've got says night transits aren't scheduled.

Lead investigator: You said you went looking for number four and couldn't find it. Talk me through that minute and a half.

Master: It's the longest minute and a half I've had. You're looking into a wall of light. The town's there, and now there's that car park with the new floodlights, and it's all one glow from out there. Four is a green flash, 0.4 of a second every two and a half. In front of that lot it's a nothing. It used to be a proper light. It was a mains light. And I'll tell you what I did without deciding to do it, which is I eased her to starboard looking for it. Five degrees. I never made that decision.

Second investigator: You've said the beacon has been getting harder to see for two years. What did you do about that, over those two years?

Master: Said it to the other masters. Said it in the tearoom. Said it to the bloke on the workboat once. Never wrote it down, never told the office, never told the port. And before you ask me why, it's because there's nowhere to put it. There's no form for the beacon looks dim. It's not our beacon. And I don't know who'd read it.

Lead investigator: The timetable changed on the first of June and the run moved into darkness. What came with that change?

Master: A new roster sheet. That's it. No brief, no plan, no conversation. The run moved an hour and twenty and the only thing anybody told me was the new departure time.

Lead investigator: You'd done twenty six of those night transits before Tuesday. How did the others go?

Master: Fine. That's the part that bothers me. Twenty six of them were fine, so by Tuesday I thought it was a normal thing to do. It isn't. It just hadn't bitten anybody yet.

Lead investigator: I have to ask about how you were feeling. Day nine of twelve, a split duty, back on at ten past one.

Master: I was tired. I'd be lying if I said otherwise. Six and a quarter hours the night before, which is about normal for me on a split. But I wasn't fighting to stay awake and I wasn't slow. I'd rather you didn't hang it on the roster, because the roster's been like that for eleven years and I knew where I was. What was different on Tuesday was the dark and the plotter.

Second investigator: After she touched, you decided not to try to back off. Take me through that.

Master: Falling tide, unknown bottom, thirty eight people aboard and I've just heard her grind for nine seconds. If I pull her off and she's holed, I'm in eleven feet of water with a list. On the bank she can't sink. That's not a clever decision, that's the first thing they teach you. Stop the engines, assess, don't make it worse.

Lead investigator: Last one from me. What do you think this investigation ought to be looking at?

Master: The beacon and the night plan. In that order. My bit I'll wear and I've already written it down. But if you fix me and you leave that channel the way it is, you've fixed nothing at all.

Interview notes, deckhand and outgoing master

The investigator's written record of two interviews that were not recorded, one at the interviewee's request and one by telephone. These notes are the record of both.

Fictional source material for the Kanooka Heads case study. People are identified by role only.

Record type: The investigator's own notes, written up the same day.

Neither of these two interviews was recorded, so these notes and the investigator's working notes beside them are the record of what was said.

Nothing here is a verbatim quotation unless it is inside quotation marks, in which case it is a phrase written down at the time because of how it was put.

1. Deckhand, 14/07/2026, 09:40, Tarrakin Ferries town wharf office

Mode: Not recorded. The deckhand asked that it not be recorded and this was agreed. A second investigator was present and has read and agreed these notes.

Free recall

Started with the changeover at the town wharf at about 13:05. Was not on the bridge for it and did not hear the handover between the two masters. Says crew changeovers happen on the aft deck while the boat is turning round and the masters do theirs in the wheelhouse.

Afternoon was ordinary. Two round trips, light loads, nothing said about the plotter or anything else.

TF18 loaded 38 at Muntari. Went round the saloons on departure doing the safety announcement and then started the fares. Was in the main saloon at the servery end for most of the transit.

Was called up to the bridge at about 18:04 and the master said to go and do the fares instead, which was normal. "He didn't want me up there and there was no reason for me to be up there."

Felt the boat change note before the alarm went, "she went heavy", and was already moving forward when the impact came. Describes the impact as a hard shove and then a long scraping.

Master was on the public address within about thirty seconds and the deckhand started the lifejackets from aft forward.

Aft saloon was easy. Forward saloon was not, because the locker is behind the kiosk counter and there were three passengers standing at it. Had to ask them to move, go round the end of the counter through a narrow gap, and hand the jackets out over the counter.

Sounded the No. 2 starboard void at the master's direction, found about 75 mm of water, and sounded it every ten minutes after that.

All 38 passengers off by workboat, last one at about 19:26.

Additional questions

Asked whether the master seemed different at any point during the transit.

Answer: no, and the deckhand would not have been able to tell, having been in the saloon from about 18:05.

Asked whether a lookout is normally kept on the bridge at night on this run.

Answer: no. "There's one bloke up there. There's always been one bloke up there." Says nobody has ever asked her to stand a lookout on any vessel in this company, in daylight or in the dark, and that the fares have to be done during the transit because there is nowhere else in the timetable for them.

Asked what she was told when the winter timetable started. Answer: the new times. Nothing else. No briefing about the run being in darkness.

Asked whether she had ever seen the No. 4 beacon. Answer: does not know which one it is. Has never been shown the channel marks and does not have a role that requires it.

Asked about the lifejacket locker: had she raised it before? Answer: no, because "it's how the boat is". Has since submitted a safety report about it (SR-2026-0291).

Follow-up questions

Still to ask: whether the other two 24 m vessels have the same server counter in front of the same locker (put to the operator; answered yes on 16/07).

Still to ask the operator: is a second person on the bridge required, or expected, or neither, for a channel transit in darkness? Nothing in OPS-3 says anything either way and the deckhand's account is that it has never happened.

Investigator's working notes

She is a careful witness and she was clear about the limits of what she saw.

She volunteered three times that she was in the saloon and could not speak to the bridge.

The lookout point is hers, not mine, and I had not put it on my list. The passage plan names a speed, a course and three beacons and says nothing at all about who is on the bridge. On a 40 m wide channel in the dark that is worth saying in the report even though I cannot carry it to a finding on this

occurrence, because the master did not fail to see anything a second pair of eyes would have seen: the beacon was screened from both of them.

The lifejacket account is the best evidence in the file for that finding and it came from a deckhand who thought it was not relevant. That is the argument for asking every witness the open question at the end.

Record not finalised at the time of drafting: the lookout question is out with the operator and her answer may need to be put to her again once it comes back.

2. Outgoing master, morning shift 07/07/2026, interviewed 17/07/2026, 11:15, by telephone

Mode: Not recorded. Telephone interview, agreed in advance, notes taken contemporaneously and read back to the interviewee at the end, who agreed them.

Free recall

Worked the morning shift on 07/07/2026, 05:40 to 13:05, four sectors.

The plotter froze twice. First time was on the 10:20 out of the town wharf, about twenty minutes in, in open water outside the entrance. Second time was on the 11:05 return, also outside the channel. Both times the display stopped responding, "the picture's there but she's not moving on it", and both times it came back by itself after a minute or two without him doing anything.

Neither freeze happened inside the Weerona Channel and he says he would have thought about it very differently if one had.

Handed over to the relieving master at the town wharf at about 13:05 and told him about both freezes. Remembers the relieving master saying it had done the same to him the month before.

Did not enter it in the defect book. Was not asked to and did not think of it as a defect because it had come good both times.

Additional questions

Asked what he would have done if it had frozen in the channel. Answer: "I'd have taken the way off her and sat there until it came back or until I was happy on the marks. There's no traffic. There's nothing to hit if you're stopped."

Asked whether there is any rule requiring that. Answer: not that he knows of. "That's just what you'd do."

Asked whether he had reported a plotter freeze at any time before. Answer: no. Believes it has happened on this vessel three or four times in the past

year, to him and to others, and that none of them has ever been written up. Asked about the No. 4 beacon. Answer, unprompted and at some length: it has been "going backwards" for two years, it is "hard work" at night now, and since the car park lighting went in it is "washed out altogether from up the top". Says he has mentioned it to the workboat crew "a couple of times" and has never reported it.

Asked why not. Answer: "Who to? It's the council's light. There's no form. You just get used to it and you look harder."

Asked whether anything accompanied the winter timetable change. Answer: a new roster and nothing else. He has done 5 of the 31 darkness transits.

Follow-up questions

Still to ask: whether the company's own reporting system has any category a master could use for an aid to navigation. Checked on 20/07: it has a "navigation hazard" category, unused on this route in three years, and no master interviewed knew it existed.

Investigator's working notes

The two masters' accounts of the handover agree in every particular, including who said what, and they agree on the thing that matters most, which is that neither of them thought a fault that had come good was something the book was for. That is not two people making the same mistake; it is the system working the way everybody in it understands it to work.

His answer on what he would have done in the channel is the most useful thing in this interview, and it also has to be handled carefully in the report. It is a hypothetical, given after the event, by a man who knows what happened.

It is evidence that the right action was available and obvious to a competent master. It is not evidence about what the relieving master would have done.

He volunteered the beacon before I raised it, which is now three of three masters who have done so.

Record finalised 20/07/2026 after the reporting-category check came back.